20251_Big Data_NoSQL(1)

nguyenductrieu.toank30chl@gmail.com Chuyển đổi tài khoản

Không được chia sẻ

* Biểu thị câu hỏi bắt buộc

Student *

Câu trả lời của bạn

Which of the following best describes the main motivation behind NoSQL databases?

- ○ To improve transaction management
- ● To simplify schema design for web-scale data
- ○ To replace relational databases entirely
- ○ To reduce server memory usage

- có improve quản lý giao dịch, vì nó nới lỏng điều kiện để hỗ trợ thực dịch dễ dàng hơn
- có giảm sử dụng bộ nhớ
nhưng ko phải mục tiêu chính

Horizontal scalability in NoSQL refers to:

- ○ Upgrading CPU and RAM of a single server
- ● Adding more machines to handle larger workloads
- ○ Replacing servers with faster disks
- ○ Compressing data to save space

vertical -> upgrading cpu or ram, thường với cơ sở dữ liệu qu

Which of the following is NOT a typical feature of NoSQL databases?

- ○ Schema-less design
- ○ High availability
- ● Complex joins
- ○ Elastic scalability

main motivation noSQL: phân tán diện rộng và tránh cần kết nối nl dữ liệu

In the CAP theorem, what does 'P' stand for?

- Performance
- Partition tolerance
- Persistence
- Parallelism

- CP: he thong han che truy cap vao cac nut ma chua duoc dong bo du lieu (ko avaiable)
- AP: he thong cho phep truy cap vao moi nut nhung du lieu co the khong giong nhau
- CA: thuong dung cho he quan tri co so du lieu quan he

According to CAP, a distributed system cannot simultaneously guarantee:

- Consistency, Availability, and Partition Tolerance
- Durability, Scalability, and Consistency
- Atomicity, Consistency, and Partition Tolerance
- Scalability, Reliability, and Performance

Which type of database is optimized for relationships and traversals?

- ○ Key-Value Store
- ○ Document Store
- ○ Column Family Store
- ● Graph Database

In MongoDB, the main data unit is:

- ○ Table
- ○ Column
- ● Document (JSON/BSON)
- ○ Node

The BASE acronym stands for:

- ○ Basic, Available, Scalable, Elastic
- ● Basically Available, Soft-state, Eventually consistent
- ○ Binary, Atomic, Secure, Efficient
- ○ Big-data, Availability, Scalability, Efficiency

Eventual consistency means that:

- dam bao nhat quan nhung sau mot khoang thoi gian nao day (noi long tinh nhat quan)

- ○ Data is always the same across replicas
- ● All replicas will become consistent over time, if no new updates occur
- ○ Each replica updates independently forever
- ○ The system never becomes consistent

The Two-Phase Commit (2PC) protocol ensures:

- ○ Availability in case of network partition
- ● Atomicity and Consistency across distributed transactions
- ○ Faster writes in distributed databases
- ○ Schema flexibility

Traditional RDBMSs scale best by adding more machines horizontally.

- ☐ True
- ☒ False

Key/Value stores like Redis and DynamoDB are ideal for complex relational joins.

- ○ true
- ● False

Document databases allow flexible schema evolution.

- (●) True
- ( ) False

Graph databases store data as nodes and edges with properties.

- (●) True
- ( ) False

BASE properties relax the strict ACID guarantees of relational systems.

- (●) True chinh nho cai eventual consistency
- ( ) False

Under CAP, if network partitions occur, you must trade off either Consistency or Availability.

- (●) True
- ( ) False

Eventual consistency guarantees that data is immediately synchronized across replicas.

- ( ) True
- (●) False

MongoDB supports both sharding and replication.

- (●) True
- ( ) False

sharding: chia data thanh cac phan manh
replication: nhan ban du lieu len de luu nhieu ban sao

Riak and DynamoDB are examples of distributed key-value databases.

- (●) True
- ( ) False

Explain briefly the difference between vertical and horizontal scaling in database systems.

Câu trả lời của bạn

Define Partition Tolerance in the CAP theorem and give a simple real-world example.

Câu trả lời của bạn

Describe how BASE differs from ACID in terms of design philosophy.

Câu trả lời của bạn

What are the advantages and limitations of schema-less design in NoSQL databases?

Câu trả lời của bạn

Why do large-scale databases such as Google Bigtable and Amazon DynamoDB often sacrifice strict consistency?

Câu trả lời của bạn

What role does replication play in achieving availability in distributed databases?

Câu trả lời của bạn

Explain why the Two-Phase Commit protocol limits scalability.

Câu trả lời của bạn

Summarize the main trade-off implied by the CAP theorem when designing distributed databases.

Câu trả lời của bạn

Gửi

Xóa hết câu trả lời

Không bao giờ gửi mật khẩu thông qua Google Biểu mẫu.

Biểu mẫu này đã được tạo ra bên trong School of Information & Communication Technology. - Liên hệ với chủ sở hữu biểu mẫu

Biểu mẫu này có vẻ đáng ngờ không? Báo cáo

Google Biểu mẫu